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Arab Open University
Faculty of Computer Studies

TM351
Data Management and Analysis

Midterm Examination (MTA)
Summer Semester 2025/2026

Time Allowed: 2 Hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

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Part 1: Multiple Choice Questions (MCQ) [10 marks]

1. The four main stages of the data pipeline are Acquisition, Preparation, Analysis and _____
   A) Archiving B) Presentation C) Encoding D) Backup
2. Which character encoding can encode Unicode characters using a variable number of bytes and is backward compatible with ASCII?
   A) ASCII B) UTF-8 C) HTML D) SQL
3. NaN stands for:
   A) New and Normal B) Not a Number C) Name and Number
   D) No Active Node
4. In table manipulation, selection means:
   A) Choosing rows B) Extracting columns C) Creating tables
   D) Removing keys
5. Which SQL aggregate function returns the average value of a numeric column?
   A) COUNT() B) SUM() C) AVG() D) MAX()
6. Which SQL clause is used to sort the result of a query?
   A) WHERE B) GROUP BY C) ORDER BY D) HAVING
7. In the relational model, a foreign key is used to:
   A) Create a chart B) Link related tables C) Delete duplicate rows
   D) Store metadata only
8. A standard database view is best described as:
   A) A virtual table based on a SELECT query B) A physical table only
   C) A backup file D) A programming loop
9. A relation is in First Normal Form (1NF) if each attribute contains only _____ values.
   A) Atomic B) Repeated C) Composite D) Missing
10. Data cleansing is mainly the process of:
    A) Detecting and correcting errors in a dataset B) Drawing diagrams only
    C) Encrypting all values D) Removing all tables

Part 2: Programming and Database Questions [40 marks]

Question One: [10 marks]

What will be the output of the following snippets of Python code? Write down the exact output, including all brackets, parentheses, braces, quotes and commas ([], (), {}, '', "", ,).
Each question = 2 marks.

1.

```
modules = ["TM351", "TM112", "M269"]
modules.append("TM354")
print(modules)
print(modules[0])
```

[2 marks]

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2

```
numbers = [5, 10, 15, 20]
print(numbers[1])
print(numbers[-1])
print(numbers[0] + numbers[2])
```

[2 marks]

3.

```
student = {"name": "Omar", "mark": 72}
student["mark"] = student["mark"] + 8
print(student["name"], student["mark"])
```

[2 marks]

4.

```
total = 1
for n in [2, 3, 4]:
    total = total * n
print("Total ", total)
```

[2 marks]

5.

```
class Course
    def __init__(self, code, hours):
        self.code = code
        self.hours = hours
c1 = Course("TM351", 8)
c1.hours = c1.hours + 2
print(c1.code)
print(c1.hours)
```

[2 marks]

Question Two: Normalization [20 marks]

The Faculty of Computer Studies at the Arab Open University needs a database to track students, academic advisors, and advised courses. The current data is stored in the following table:

Student_ID	Student_Name	Advisor_ID	Advisor_Name	Course_Code	Course_Title	Advising_Status
S01	Ahmed Ali	A01	Dr. Rafeek	TM111	Introduction to Computing	Approved
S01	Ahmed Ali	A01	Dr. Rafeek	M251	Object-Oriented Programming	Pending
S02	Mona Said	A02	Dr. Sara	TM112	Introduction to Computing and IT 2	Approved
S03	Omar Hassan	A01	Dr. Rafeek	TM351	Data Management and Analysis	Approved
S03	Omar Hassan	A01	Dr. Rafeek	TM354	Software Engineering	Pending

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The relation can be represented as:
FCS_ADVISING(Student_ID, Student_Name, Advisor_ID, Advisor_Name, Course_Code, Course_Title, Advising_Status)

Assume the following functional dependencies:

Student_ID→Student_Name,Advisor_ID
Advisor_ID→Advisor_Name
Course_Code→Course_Title
Student_ID, Course_Code → Advising_Status

- **Task 1:** Normalize the data into 1NF, 2NF, and 3NF. Show the relations for each form. **[15 Marks]**
- **Task 2:** For the final 3NF relations, identify the Primary Keys and Foreign Keys using the following format: **Table Name | Primary Key | Foreign Key**. **[5 Marks]**

Question Three: SQL Queries [10 marks]

(A) : Consider the following schema:

SUPPLIER (supplier_id, supplier_name, city)
Primary Key: supplier_id
ITEM (item_id, item_name, category, price)
Primary Key: item_id
SUPPLIES (supplier_id, item_id, quantity, supply_date)
Primary Key: (supplier_id, item_id, supply_date)
Foreign Key
supplier_id references SUPPLIER(supplier_id)
item_id references ITEM(item_id)

Write SQL queries for the following:

1. List the item names and prices of all items in the 'Stationery' category with price < 100. [2 Marks]
2. List the supplier names and item names for all supplied items. Use JOIN. [3 Marks]
3. Update the price of all items in the 'Electronics' category by increasing it by 15%. [2 Marks]
4. Calculate the total quantity supplied for each item. [3 Marks]

Part 3: Data Format Questions [10 marks]

Question One: Convert XML to JSON [5 marks]

```
<company>
 <employee>
  <id>E501</id>
  <name>Huda</name>
  <department>IT</department>
 </employee>
 <skills>
  <skill>SQL</skill>
  <skill>Python</skill>
 </skills>
</company>
```


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Question Two: Convert JSON to XML [5 marks]

```
{
  "delivery": {
    "deliveryID": "D-77",
    "customer": "Huda",
    "items": {
      "item": [
        "Book",
        "Pen"
      ]
    },
    "address": {
      "city": "Muscat",
      "area": "Seeb"
    }
  }
}
```

End of questio



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